Homework #4

This is the homework for week 4. The due date is 11:59 p.m., October 7. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (50 points)

The attendance data is presented in the following table:

Time stamp	t		attendance
9/10	1	X_1	85
9/12	2	X_2	73
9/17	NA	NA	NA
9/19	3	X_3	68
9/24	4	X_4	56
9/26	5	X_5	43

(a) Suppose $X_1, X_2, ..., X_5 \overset{iid}{\sim} Pois(\lambda_0)$.

i. Find the maximum likelihood estimate for λ_0.

ii. Suppose the enrollment is 120, find the maximum likelihood estimate for the attendance rate $\lambda_0/120$.

(b) Suppose the statistician is interested in the attendance rate, then s/he chooses to work with the model $X_i \overset{ind}{\sim} Bin(n_i, \pi_0)$ for $i = 1, 2, ...$, where the enrollment $n_1 = n_2 = 100$ and $n_i = 120$ for $i = 3, 4, 5$.

i. Find the maximum likelihood estimator for the attendance rate π_0.

ii. Find the maximum likelihood estimate for the attendance rate π_0.

(c) There seems a declining trend in the attendance data: fewer students took the attendance survey. To estimate the time trend, the modeller works with the following model:

$$X_t \overset{IND}{\sim} Pois(\alpha + \beta t) \text{ for } t = 1, 2, ..., 5.$$

Write down the log likelihood function in terms of α and β.

(**Question2):** (30 points)

(a) Suppose $X_1, ..., X_n \overset{iid}{\sim} Unif(0, \theta)$, i.e., the density

$$f(x) = \begin{cases} \frac{1}{\theta} \text{ if } x \in (0, \theta), \\ 0, \text{ otherwise.} \end{cases}$$

We already know that the maximum likelihood estimator of θ is $\hat{\theta} = \max(\{X_i\}_{i=1}^n)$. Derive the CDF and the pdf of $\hat{\theta}$. (Hint: Use the fact that $Prob(\max_{i=1}^n X_i \leq x) = Prob(X_i \leq x \text{ for all } i)$ and the density is the derivative of the cdf.)

(b) Suppose $Y_i \overset{iid}{\sim} Exp(\lambda)$ for $i = 1, ..., n$, i.e., the density

$$f(x) = \lambda \exp(-\lambda x), x \geq 0.$$

Find the maximum likelihood estimator for λ, and derive its sampling distribution. (Hint: if $X \sim Gamma(a, b)$ under the shape-rate parameterization, then $1/X$ follows the inverse Gamma distribution (see the wiki link) : $1/X \sim inv - Gamma(a, b)$.)

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

i. Find the maximum likelihood estimate of Q1 (c) numerically.

ii. Simulate $Y_i \overset{iid}{\sim} Exp(\lambda)$ for $i = 1, ..., 100$, we choose $\lambda = 2$ in this homework. With the pivotal quantity method, compute a 90% CI for λ. Does your 90% CI contain the true value?

iii. Repeat the experiment of ii for 1000 times. Calculate the coverage rate of your CI.